Thinking Like a Scientist

Science begins with curiosity. The skill is turning an everyday “Why?” into a question you can actually investigate.

What investigation in science means

Science doesn't start in a fancy laboratory — it starts with **curiosity**. Even your kitchen is a place to do experiments. The textbook's example: *why does a puri puff up, and why is one side thinner than the other?* A scientist turns that wonder into a careful investigation.

> **The big idea —** we don't just want to *learn new facts*, we want to learn *how to find new facts*. That process is called **systematic investigation**.

The steps of a scientific investigation

1 **Ask a scientific question** — one you can test by observing or experimenting.

2 **Decide what you can change** (thickness of dough, type of flour, oil temperature).

3 **Decide what you can observe or measure** (Did it puff? Yes/No. How long? In seconds).

4 **Experiment, changing only one thing at a time**, and keep careful notes.

5 **Use your observations** to improve your understanding — which usually leads to new questions.

Observation vs Inference

Observation	Inference
What you directly notice with senses or instruments	An explanation you *work out* from observations
"The puri puffed up; one side is thinner."	"The thin side probably trapped steam that pushed it up."

Two kinds of observation

Qualitative (quality)	Quantitative (quantity)
In words / yes-no — "the oil smoked."	Number + unit — "it puffed in 8 seconds."

> **Symbols to remember:** the **root** = stay grounded in real observation; the **kite** = let curiosity take flight. Good science balances both.

Where students lose marks

> **Trap 1 — Mixing observation and inference.** "It puffed because of steam" is an *inference*, not an observation. An observation is only what you directly sensed or measured.

Trap 2 — A vague question is not scientific. "Are puris nice?" can't be tested. "Does thicker dough take longer to puff?" can.

Trap 3 — Forgetting the unit. A quantitative observation needs a number *and* a unit. "8" is incomplete; "8 seconds" is correct.

Slip — thinking experiments need special equipment. Everyday things are enough; the careful method is what matters.

Model answers — write them like this

Example A. Classify: "The ice cube became smaller because the room is warm."

Step 1: "The ice cube became smaller" = directly seen → observation.

Step 2: "because the room is warm" = a reason worked out → inference.

∴ It contains both an observation and an inference.

Example B. Turn "Why are some puris better?" into a scientific question.

Step 1: Pick one thing to change (oil temperature) and one to measure (time to puff).

Step 2: Phrase it so an experiment can answer it.

∴ "Does hotter oil make a puri puff up in less time?"

Example C. Give one qualitative and one quantitative observation while frying a puri.

Step 1: Qualitative = words/yes-no. Quantitative = number + unit.

∴ Qualitative: "the puri puffed up." Quantitative: "it took 8 seconds to puff."

Ready to practise? → Open the worksheet